

A Model of the Perceptual and Conceptual Processes in Graph Comprehension

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The article proposes that graph comprehension emerges from an integrated sequence of several types of processes: pattern-recognition processes that encode graphic patterns, interpretive processes that operate on those patterns to retrieve or construct qualitative and quantitative meanings, and integrative processes that relate these meanings to the referents inferred from labels and titles. The model is supported by 2 studies that examine the pattern and durations of eye fixations as a person interprets line graphs or answers questions about line graphs that vary in type and complexity. One implication is that graph comprehension might be more accurate and more complete if the graph's format were changed or the audience were educated to lessen the burden of the inferential, interpretive, and integrative processes.

In this article, we propose a model of the comprehension of line graphs that emphasizes the close interaction between conceptual processes, such as interpreting labels and scales, and perceptual processes, such as encoding and interpreting the slopes and patterns of the lines themselves. The model accounts for the pattern and duration of the eye fixations of college students who are interpreting line graphs with three dimensions. Moreover, it provides a theoretical framework for analyzing the processing of more complex graphs, more complex tasks, and other types of audiences (novices vs. experts). This focus on the conceptual aspects of graph interpretation contrasts with the implicit assumption of some researchers and graphic designers that graph comprehension is

primarily or solely pattern recognition. Indeed, evidence that graph comprehension processes involve much more than pattern recognition emerges from task analyses of graph comprehension (Bertin, 1983) and from empirical studies that document the numerous errors that are made when people interpret even simple graphs (Culbertson & Powers, 1959; Guthrie, Weber, & Kimmerly, 1993; Romberg, Fennema, & Carpenter, 1993; Vernon, 1946, 1950). Also, college students and even graduate students with significant research experience can have a remarkably incomplete understanding of the functional relations in relatively simple line graphs (Shah & Carpenter, 1995). Finally, the time scale for graph comprehension is consistent with the execution of complex cognitive processes. The spontaneous description of a typical line graph in the present experiments takes approximately half a minute, which is comparable to the time needed to comprehend a paragraph of moderate length and two orders of magnitude greater than the 300–500 ms that it takes to recognize a pattern, such as a face, a word, or an object.

Task Analysis

The present research builds on a task analysis of graph interpretation that was first described by

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Bertin (1983) and modified by other researchers (e.g., Cleveland & McGill, 1985; Guthrie et al., 1993; Lohse, 1993; Pinker, 1990; Simkin & Hastie, 1987). The analysis suggests that there are three major types of processes in graph comprehension: encoding the visual patterns of the display, translating the depicted patterns into the associated conceptual or quantitative relations, and identifying the referents of the relations. We will describe the content of these processes and provide a model of how they are organized.

As described earlier, pattern recognition has been the focus of most of the research on graph comprehension (e.g., Cleveland & McGill, 1984, 1985). Although there is no doubt that pattern recognition is an important component of graph comprehension, in the present model, it is simply one component. The larger contribution of the present study is to begin to specify what features constitute a visual chunk or pattern in line graphs that are being interpreted by quantitatively oriented college students. In this context, a pattern may correspond to a distinct x - y function. *Distinct* functions are lines with different slopes indicating qualitatively different x - y relations. For example, a line with a positive slope is distinct from one with a negative slope. By contrast, we will show that parallel lines contribute only a little additional complexity to the graph comprehension.

A second process is the translation of visual features into the conceptual relations that are represented by those features (Kosslyn, 1989; Pinker, 1990). The translation may involve retrieving the quantitative interpretation associated with the visual pattern, such as the knowledge that an upwardly curved line represents an accelerating function. However, if the visual features have not been previously associated with quantitative interpretations, the person must infer or reason out the quantitative or qualitative interpretation. Also, the translation process can be complicated by the existence of multiple graphic features, such as several x - y lines in a line graph, that must be integrated into a unified interpretation. Thus, the effort required to interpret a graphic pattern depends partly on the pattern's familiarity and on the number of distinct patterns to be interpreted and integrated.

A third major process includes determining the

referent of the concepts being quantified and associating those referents to the functions (Bertin, 1983). This step typically requires inferring the referents from the labels, title, and caption, as well as from whatever keys and symbols are intrinsic to the display. Although this process has received little empirical attention, there have been several design suggestions for minimizing its difficulty (Kosslyn, 1994; Schmid, 1983).

Although these three types of component processes have been described in several task analyses of graph interpretation, the analyses leave unspecified the temporal dynamics and organization of the processes. The implicit assumption might be that the three processes are executed in sequentially ordered steps. For convenience, we will refer to this default model as the *pattern-recognition model*, recognizing the primacy afforded the proposed first step. In this model, comprehension occurs by the person first encoding the graphic pattern in toto, then translating the pattern into its quantitative interpretation, and finally, relating that information to the variable names and numerical information in the scales. The assumption is that most of the cognitive action comes in the first step, recognizing the graphic pattern, even though pattern recognition must be followed by other steps.

A second model, which we shall call the *integrative model*, provides a more complex view of graph comprehension. In this model, pattern-recognition processes provide the visual chunks that are digested by interpretive cognitive processes. Comprehension entails encoding a visual pattern, translating that pattern into its quantitative and qualitative interpretation, and relating that information to the referents that are inferred from the variable's name, the scales, and specific values. This recognition and interpretation cycle is repeated with different and more complex visual chunks to build a coherent interpretation. Indeed, in Experiment 2, we show that the number of gazes on a line graph is highly correlated with the number of distinct line functions, each of which constitutes a visual chunk in the context of a three-variable line graph. The main point is that graph comprehension is an incremental process that involves pattern recognition, interpretation, and integration. Moreover, if the complexity of the graph increases, the integrative model suggests that the cycle of processes

would scale up as a function of the additional chunks to encode and interrelate. Similarly, if a person were less familiar with the graphic conventions, then the comprehension process should have more cycles, reflecting the smaller chunks that would be recognized per cycle and the larger amount of problem solving in the interpretation of those chunks. The organization that emerges in comprehending a line graph of multiple visual "chunks" would be serial, reflecting the successive cycles of encoding, interpretation, and integration, even though the entire graph is visually accessible and at least some visual features may be encoded in parallel. Such an organization is consistent with other models of display-based problem solving that have emerged from studies of people's eye fixations and verbal reports, such as models to account for the comprehension of a simple mechanical device from a diagram (Hegarty, Carpenter, & Just, 1991); the solution of a complex visual analogy (Carpenter, Just, & Shell, 1990); or the mental rotation of one figure into another (Just & Carpenter, 1976, 1985).

To distinguish between the pattern-recognition model and the integrative model, we examined the patterns of gazes of college students who described or answered questions about line graphs that varied in complexity. A *gaze* refers to a sequence of consecutive eye fixations on a display region that is meaningful in the model, unbroken by fixations on other regions. The regions in the current model correspond to large segments of the visual display: the main graphic pattern, the title, the x label, the y label, and the z label. In these tasks, as in many other display-based tasks, the person tends to fixate the referent of the symbol being mentally operated on (Just & Carpenter, 1976, 1980). Because of this association between the locus of the gaze and the underlying cognitive process, the analyses of eye fixations have been useful for enriching the theoretical description of language-comprehension processes (Just & Carpenter, 1980), complex problem solving (Carpenter et al., 1990), and visualization (Just & Carpenter, 1985). Although people obviously can dissociate their locus of gaze from the focus of their cognitive processing, such dissociation is not encouraged by the structure of the current task.

Graphic Complexity

The present studies involve line graphs of three continuous variables, which is a commonly used format for representing data in the social and physical sciences. Three-variable data sets are useful in this research because they allow a wide range of complexity, including interactions. In addition, such data sets can be graphically depicted in two different ways because there is a choice of which of the two nondependent variables to assign to the x axis and which should be a parameter on the curve (the z -variable). Thus, it is possible to maintain the same data and yet systematically vary the visual properties of the graph in the same format.

In previous studies, we found support for the hypothesis that distinctive x - y functions are units of encoding and interpretation (Shah & Carpenter, 1995). For example, the graphs in Figures 1a and 1b often elicit different descriptions. These include a limited number of propositions that describe the quantitative concepts that are associated with the x - y functions, concepts such as "linear" or "increasing." People rarely spontaneously describe the metric properties of the z - y relations. Because of this aspect of graph interpretation, two alternative perspectives of the same three-variable data set can be psychologically non-equivalent. Indeed, the two perspectives of the same data set, such as in Figures 1a and 1b, are often judged to represent incompatible data in a sequential comparison task.

In the current studies, we pursue the processing consequences of this divergence between the alternative perspectives of the same data set. Specifically, if two perspectives of the same data set depict different numbers of distinct x - y functions, then the two perspectives may take different amounts of time to interpret. Consider, for example, the cross-over perspective of the interaction in Figure 1a. An important property of this graph is that the x - y functions for both values of z are increasing in slope. According to our predictions, the person is likely to code the functional relations similarly and have little understanding of the z - y relation. By contrast, in the non-cross-over perspective in Figure 1b, the slopes of the x - y lines reverse for the two values of z (age, in this case), and the reader is expected to encode two different quantitative functions and

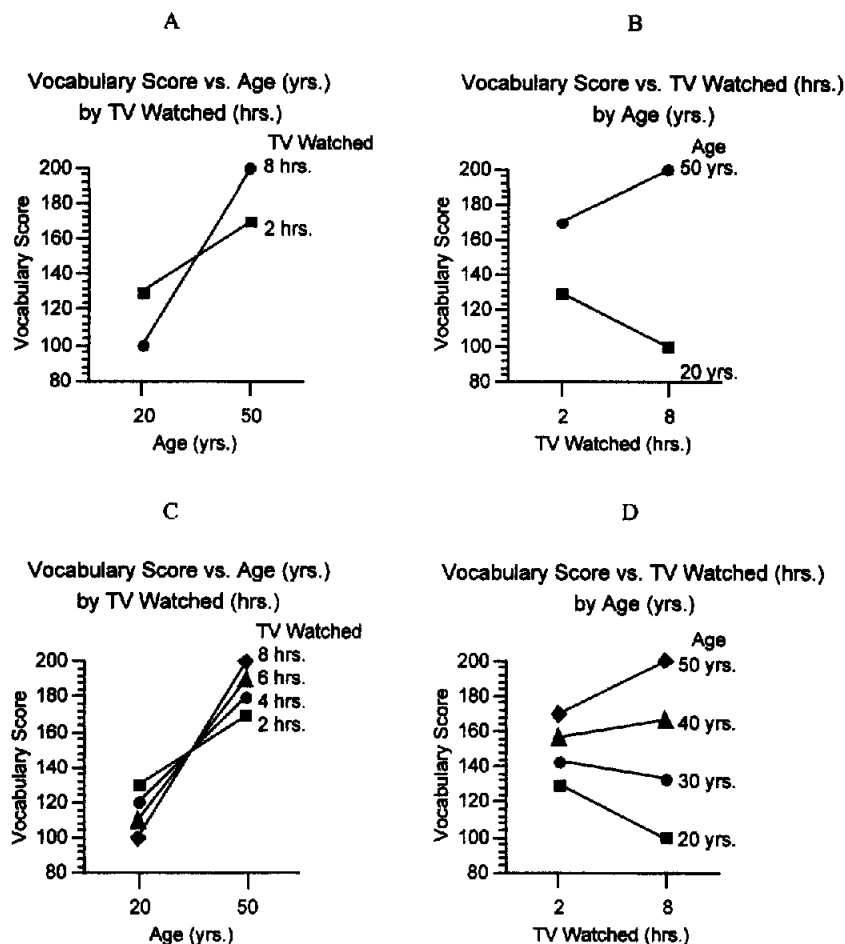


Figure 1. Two perspectives of the sparse and dense versions of the cross-over interaction data set that is pictured in Figure 4. Figure 1a and 1c are the sparse and dense versions of the cross-over perspective, and Figure 1b and 1d are the sparse and dense versions of the non-cross-over perspective.

have a more detailed understanding of how the z -variable relates to the y -variable. Consequently, the non-cross-over perspective is predicted to take longer to encode than the cross-over perspective.

A second way in which we manipulated the complexity of the graphic pattern was by varying the number of lines in a graph. Specifically, we interpolated additional x - y lines at equal z -intervals between the x - y lines of the sparse version of a data set in order to make a dense version. Figures 1c and 1d present the dense versions of the functions in Figures 1a and 1b, respectively. In each case, additional lines are interpolated between the extremes established by the x - y lines in Figures 1c and 1d. Because the

dense data sets contain more lines, they might be expected to be more psychologically complex. However, the proposed model predicts a more complex pattern of results: The effect of the density should depend on the particular x - y pattern. If the additional lines in the dense data sets are similar in direction and slope (as in Figure 1c), the density manipulation should not greatly increase the psychological complexity of the graph, and there should be little effect on the interpretation time or number of gazes. By contrast, if the additional lines are qualitatively dissimilar, as in Figure 1d, then the manipulation increases the number of distinct functions. If each distinctive function initiates a cycle of encoding, interpretation, and integration, then an increase in

the number of different functions should increase the time and number of gazes executed during the interpretation of a graph.

The density of the data set may also influence the kinds of interpretations that people give to the z - y relations, in addition to the x - y relations. Specifically, increasing the number of lines on a graph may increase the perceptual salience of the z - y information. Thus, at some point, a richer, more complex data set may facilitate the interpretation of the z - y relations. In summary, the graphs were designed so that different internal representations would tend to be evoked by the two perspectives of the same underlying data set. In this section, we describe the three types of data sets that are used in the present studies and the predicted effect of interpolating lines to create a "dense" version. Overall, the model predicts that the graphic density would influence the interpretation of those graphs for which additional lines result in additional, distinct functions. Hence, the effect of density should vary with the data set and perspective.

Cross-Over Interaction Data Set: Cross-Over and Non-Cross-Over Perspectives

In this data set, illustrated in Figures 1a and 1b, one variable has a greater or lesser effect, depending on the value of the second variable. However, the two alternative perspectives of this data set can result in different representations because the graphs differ in which aspects of the interaction they make salient through the slopes of the depicted x - y lines. In the cross-over graph in Figure 1a and 1c, the salient feature is the main x - y relation, showing in this case that both functions increase with the x -dimension, although at different rates. By contrast, in the non-cross-over perspective of the same data points (as in Figure 1b and 1d), the salient x - y relations show lines that have the opposite slope, indicating that the direction of the effect of x on y is different depending on the value of z . This graph (the non-cross-over) is expected to result in a more complex internal representation of the relations in the data set, one that is more likely to explicitly represent the z -variable as one that interacts with the x -variable. Now consider the effect of the density manipulation. The additional

lines in the dense cross-over perspective are expected to have little effect on the time and number of gazes because the slopes of the additional functions are similar to those in the sparse version. By contrast, in the non-cross-over data set, the additional lines have slopes that are intermediate between the opposite slopes of the extremes, and so they should evoke some additional encoding and interpretation processes.

Acceleration Data Set: Curve and Intercurve Spacing Perspectives

This type of data set contains an accelerating (or decelerating) function and a linear function. In one perspective, the decelerating (or accelerating) function is realized as a curve on the x -axis, and hence, the curve perspective, as in Figure 2a. By contrast, when plotted on the z -axis, the deceleration (or acceleration) is manifested as unequal spaces (y -increments) between successive lines that represent equal increments in the z -dimension. This intercurve spacing perspective is exemplified in Figure 2b. The decelerating function may be less accurately encoded when it is plotted on the z -axis (as in Figure 2b), as in the intercurve spacing perspective, than when it is plotted in the curve perspective (as in Figure 2a) because the inequality of the successive spaces between lines may be less visually salient than the acceleration in the curve perspective. Also, if people are less familiar with interpreting a change in intercurve spacing as representing a decelerating (or accelerating) mathematical function, then more effortful inferences may be required to construct the interpretation of this pattern. With respect to the density manipulation, additional interpolated lines should have relatively little effect on the times for the curve and intercurve spacing perspectives because for both perspectives, the additional lines are parallel to the existing lines.

Common Value Interaction Data Set: Line and Point Perspectives

In this type of interaction, one independent variable has no effect on the y -variable for one value of the second independent variable. However, the lack of effect is represented differently for the two perspectives and may result in

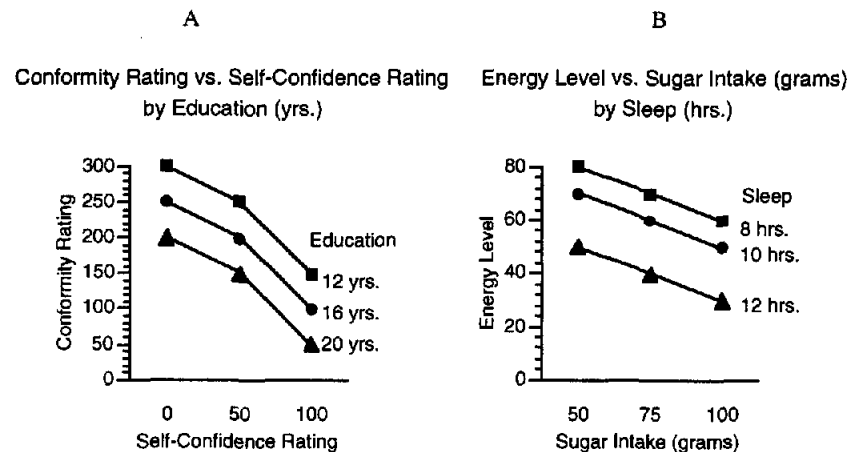


Figure 2. Two perspectives of an exemplar used to investigate the linearity assumption in encoding the third variable. The graph in Figure 2a is an example of a curve graph, in which the x -variable is curvilinear, and hence the nonlinearity is salient and familiar. Figure 2b is an example of an intercurve spacing perspective, in which the nonlinearity in the z -variable can be inferred from the unequal y -spacing between equally spaced values of the z -variable (in this case, hours of sleep).

different internal representations. For the line perspective (see Figure 3a), one of the salient properties is the horizontal line, which may automatically evoke the retrieval of the associated conceptual relation "no effect." For the point perspective (see Figure 3b), the salient properties may include the functions with increasing or decreasing slopes but may not include the lack of effect, which is depicted by the two (or more) functions meeting at a point. Hence, the

lack of effect may be more likely to be part of the interpretation of the line perspective than of the point perspective. For the common-value interaction data set, the effect of the density manipulation again should depend on the perspective. For the point perspective (e.g., Figure 3b), the interpolated lines have approximately the same slope, so additional lines should not greatly complicate the interpretation. By contrast, in the case of the line perspective (e.g., Figure 3a), the interpolated lines

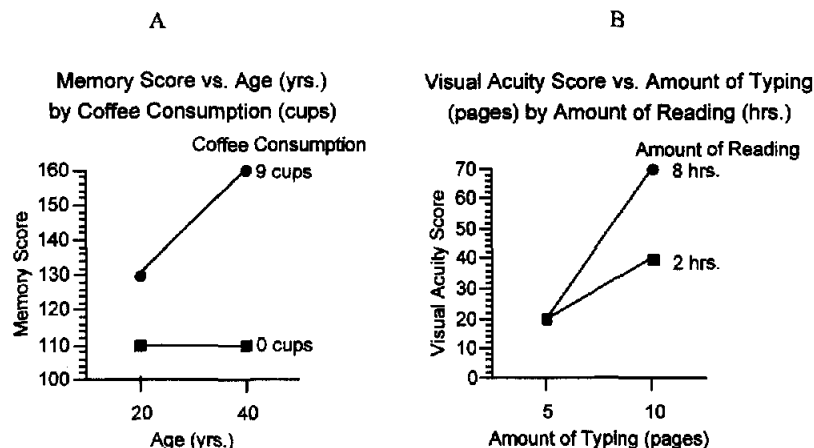


Figure 3. Two perspectives of a data set that depicts an interaction in which there is no effect of one of the variables at one value of the other variable. Figure 3a shows the line perspective, in which this lack of effect is more salient than the point perspective, Figure 3b.

bridge between a flat line and a steeply sloped line and may look like an intermediate category, neither flat nor steeply sloped. Consequently, the manipulation was expected to affect the time and number of gazes, depending on the perspective.

To develop a more precise characterization of graph comprehension, the present research traces the processes through the pattern of gazes and verbal descriptions that participants provide in order to describe (in Experiment 1) or answer questions about (in Experiment 2) graphs that vary in type and complexity.

Experiment 1

This experiment examines the pattern of gazes on line graphs and their descriptions, taking as its starting point the contrasting predictions made by the two models of how the basic processes are organized: the pattern-recognition model and the integrative model. These models generate different predictions about the distribution of gazes in different sectors of the graphs as well as the effect of the graph's characteristics on the comprehension processes.

The pattern-recognition model makes two predictions about the distribution of time across different sectors of the graph. The first is that the amount of time or the number of gazes on the axes and labels is small, perhaps only one gaze that lasts whatever time is required to identify the variable's name and the range of values. By contrast, the time spent on the sector with the main pattern should be a relatively large proportion of the total time. A second prediction is that the complexity of the data set should influence the time on the main pattern but not necessarily the time on any other component, such as the labels. This prediction emerges because it is expected to take longer to encode the pattern. However, if the variables themselves need only be identified once, then the complexity of the data set should have little effect on the number or duration of inspections on the variables' names or axes. A slightly more complex version of this hypothesis assumes that each line's label may require independent identification of the *z*-legend. In that case, the number of lines should influence the time and number of gazes on the *z*-legend and values but not the time on the other axes and labels. In either case, the time on the

x-axis and *y*-axis should be minimal and not vary according to the density manipulation.

The integrative model proposes that comprehension involves cycles of encoding and interpretation through the labels and legend. This model predicts that a large proportion of the gazes and the gaze time should occur on the regions that are used to encode referents, namely the *x*-axis, *y*-axis, and *z*-legend, relative to the time on the main pattern. In addition, an increase in the number of distinct *x*-*y* functions in the graph should result in an increase in the time on the main pattern and also the time spent on the *x*-axis, the *y*-axis, and the *z*-legend.

To examine the graph-comprehension processes, the eye fixations of participants were monitored while they provided a verbal description of a three-variable line graph. The description task was followed by a perspective-comparison task in which the first graph was taken away and the person was shown a second graph. The task was to judge whether the second graph depicted the same data set from an alternative perspective or a different data set.

We predicted that the patterns of eye fixations would support the integrative model of graph comprehension. In addition, we expected that the person's internal representations of data would be limited in the same way that they were in our previous studies of line-graph comprehension. Specifically, we expected that both the verbal descriptions and comparison task would provide evidence for the asymmetry in the representation of alternative perspectives due to the emphasis on *x*-*y* relations and the deemphasis on *z*-*y* relations.

Method

Task. Each participant examined an initial graph in order to briefly describe it while her or his eye fixations were monitored remotely. Then the initial graph was removed and a second graph appeared that had to be judged as depicting either incompatible data or compatible data (actually, identical data from a different perspective). Each person judged a total of 12 pairs of graphs, and our primary interest was the interpretation of the initial graph of each of the 12 pairs. The comparison task provided converging evidence for the person's interpretation. In addition, it helped to convey to the participant a specific sense of what

it meant to interpret a graph (i.e., sufficiently to judge the relations of the first graph to those shown from a different perspective).

Participants. Thirty-four Carnegie Mellon students participated for course credit or payment. All were native English speakers and had normal or corrected-to-normal vision. The data for an additional 6 people could not be analyzed because of an inability to accurately record their eye fixations.

Design. This experiment used a mixed design in which there were two within-subject factors—the three types of data spaces and the graphic perspective—and a between-subjects factor—the density of the data set. Participants were randomly assigned to view either sparse or dense graphs. There were two stimulus presentation orders: one that was initially randomized, subject to the constraint that two graphs of the same type of data set could not directly follow each other, and another that was simply the reverse of the initial randomization. Participants were assigned to one of the two orders.

Stimuli. Figures 1–3 are examples of the initial graphs that were viewed by participants. There was a total of 24 pairs of experimental graph pairs (initial and comparison), although only 12 pairs (sparse or dense) were shown to any one person. The 24 initial graphs were constructed in the following manner. First, two exemplars were generated for each of the three data sets. The two exemplars were intended to be minor but distinguishable variants representing the same qualitative relations. For example, for the cross-over interaction data, one exemplar had two functions (in the cross-over perspective) that were steeply increasing, whereas the other exemplar had two functions that were both steeply decreasing. Next, we generated the two perspectives of the six data sets, resulting in 12 graphs. For each graph, the sparse version was generated using only two or three x - y lines, whereas the dense version was generated by interpolating two or three additional lines between those in the sparse version, resulting in 24 graphs.

Each graph was randomly assigned a set of three names that had been chosen so that the underlying factors might plausibly be related to one another (e.g., vocabulary score, age, and TV watched), but the relation is not commonly known. This aspect of the design was intended to

minimize the influence of the participants' knowledge about specific relations while providing labels that would be easily understood. The fact that there were no overall, systematic differences between different exemplars of the same graph perspective suggests that the content of the variable names did not have a major influence in this study.

For the comparison task, a graph was designed from each initial graph by switching the x - and z -variables. For the "same" condition (half of the trials), the participants saw the identical data points plotted from the alternative perspective. For the "different" condition (the other half of the trials), the data in the alternative perspective were altered to substantially change the metric properties of the z - y relations of the initial graph.

Each graph appeared in the center of the screen. As illustrated by the graphs in Figures 1–3, the x - and y -variables were clearly labeled near the axes, and the values of the z -variable were labeled next to the individual lines and depicted the values of that variable. Finally, a title above the graph specified all three variables, following the conventional order of mention: y as a function of x by z .

Procedure. Before the experiment began, the participants were familiarized with examples of line graphs and with the task of describing and comparing graphs. In addition, they were given instructions about the tracker and familiarized with the equipment. In none of the examples did the experimenter tell the participant what kind of detail to provide in describing a graph.

Once a trial began, the participant read the title of the graph and then described the initial graph. Thus, the inspection pattern and durations include the time that the participant took to read the title and describe the initial graph. After they described the initial graph, they pushed a button to terminate its presentation and to view the comparison graph. Participants were told not to give detailed attention to specific values in the comparison task, and they were told that the two graphs could be different perspectives of the same data set. The experimental session included 1 pair of practice graphs, 12 experimental pairs, and 5 interspersed filler pairs that showed the same x - y patterns in both perspectives (e.g., parallel lines with positive slopes).

Several features of the experiment were de-

signed to instruct and remind participants that the z -variable of the initial graph was continuous, rather than nominal or ordinal, so that a failure to encode the metric properties of the z -variable would not reflect the incorrect inference that the z -variables were ordinal or nominal. The continuous nature of the z -variable was described in the instructions. Moreover, this property was implicit in the types of z -variables, which always referred to continuous dimensions (e.g., amount, not type); in the quantitative values in the legends, which were never nominal; values in the legends; and in the perspective-comparison task itself, in which the x - and z -variables were interchanged.

Eye fixations. The participants' eye fixations were measured using an Iscan (Cambridge, MA) Model RK-426 corneal-reflectance and pupil-center eye tracker. The tracker sampled the position of the participants' gaze every 16.7 ms and output the x and y coordinates to a Digital Equipment Corporation (Nashua, NH) Vaxstation 3200. The individual x - y coordinates were aggregated into fixations. For the analysis, the graph was divided into five regions: the pattern, the x -axis (including values and labels), the y -axis, the z -labels (and values), and the title, as shown in Figure 4. The regions were defined in the same way for all of the graphs. A graph subtended

approximately $19^\circ \times 14^\circ$, and the tracker is accurate to better than 1° of visual angle.

Successive fixations in the same region were aggregated into units called gazes, which constituted the main eye-fixation measure. Only gazes that were longer than 250 ms were included in the analysis, a criterion that was intended to minimize the effects of noise, short fixations, saccades, and blinks. With this criterion, an average of 10.7% of the total gaze duration was not assigned to a region. The proportion of missing gazes did not significantly vary as a function of complexity or perspectives for the common-value data set. For the remaining data sets, the proportions varied by only a small amount: 10.0% of the total duration for the cross-over perspective, 11.8% for the non-cross-over perspective, 9.5% for the curve perspective, and 10.8% for the intercurve spacing graph. A total of 30 trials from 16 different participants were not analyzed for several reasons: (a) a total of more than 15% of the observations in that trial were unclassifiable (due to blinks, noise, or lost observations—16 trials); (b) the trial was longer than 75 s (8 trials); or (c) the trial was inadvertently skipped by the participant (6 trials). The missing data were replaced by estimating the observations based on the total mean, the participant mean,

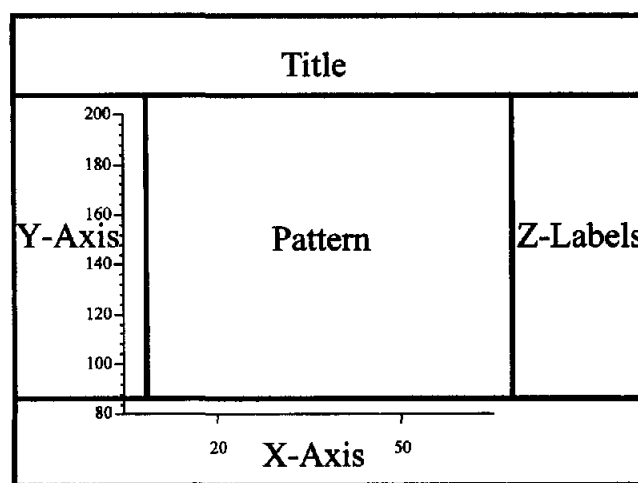


Figure 4. The five regions of the display for the analyses of gaze durations: pattern, x -axis, y -axis, z -labels, and title. The pattern region contains the data points and lines. The x -axis and y -axis region includes the x -axis and label and the y -axis and label, respectively. The z -labels include the values and units associated with the graph lines. Finally, the title provides the names of the variables, and potentially, information about what the units of a variable represent. The regions were defined identically for all the different graphs.

and trial-type mean, using a procedure in Winer (1962).

Data analysis. Table 1 presents the total gaze durations and the durations on individual sectors for each of the three data sets because our main predictions concern the effects of perspective and density for the four graph types within a data set. An analysis of variance (ANOVA) on each of these measures assessed the significance of the manipulations of perspective, density, and their interaction for each data set. Table 1 reports any *F* values that were larger than 2.5 and indicates the significant effects for the duration measures, or secondarily, the number-of-gazes measures. Although the multiple analyses increase the likelihood of finding significance by chance, the convergence of major results between this experi-

ment and the next study increases our confidence in the major findings.

Classifying verbal descriptions. The participants' descriptions of the initial graph were classified according to the information given about the *z*-variable, specifically, whether it was described (a) nominally, (b) ordinally, (c) metrically, or (d) not mentioned at all.

1. Nominal descriptions included the names of the *z*-values but did not provide any ordinal or metric information about the *z*-*y* relation. For example, for the *non-cross-over* perspective graph (shown in Figure 1b) that related hours of TV watched (the *x*-variable) and noise level (the *z*-variable) to vocabulary score (the *y*-variable), a nominal description would be, "For age 20, an increase in the amount of TV watched shows a

Table 1
*Mean Viewing Time (in Seconds) on Various Sectors
and Large F Values (Experiment 1)*

Graph type	Total	Pattern	X-axis	Z-label	Y-axis	Title
The cross-over interaction data set						
Non-cross-over, sparse	34.00	7.00	6.40	4.97	6.20	5.05
Non-cross-over, dense	40.61	9.78	5.53	7.77	7.19	5.51
Cross-over, sparse	35.66	9.42	6.28	5.19	5.71	5.33
Cross-over, dense	35.87	12.01	5.06	5.89	4.73	4.69
Perspective	—	18.74**	—	—	14.42**§§	—
Density	—	4.30*	—	2.41§	—	—
Perspective × Density	5.82*§§	<1§	—	3.89§	6.46*§§	4.28
The common-value interaction data set						
Line, sparse	33.52	6.26	7.24	4.49	6.30	4.89
Line, dense	39.05	8.42	7.24	7.28	6.64	5.33
Point, sparse	34.70	7.57	6.89	5.42	5.91	5.42
Point, dense	36.11	7.08	6.63	6.44	6.34	5.84
Perspective	—	—	—	—	<1§§	—
Density	<1§	—	—	4.32*	—	—
Perspective × Density	2.58§	4.65*	—	6.42*§§	—	—
The acceleration data set						
Curve, sparse	34.58	10.91	6.20	4.82	4.76	4.71
Curve, dense	36.01	10.76	5.87	5.09	5.73	5.06
Intercurve, sparse	32.97	8.84	5.48	5.75	5.02	4.33
Intercurve, dense	32.43	7.80	4.70	7.26	4.59	4.66
Perspective	4.24*	18.17**	4.18*§	13.40**§§	—	—

Note. Dashes indicate that the associated *F*-statistic was small.

p* < .05, gaze-duration effect significant. *p* < .01, gaze-duration effect significant.

§*p* < .05, number-of-gazes effect significant. §§*p* < .01, number-of-gazes effect significant.

decrease in vocabulary score, but for age 50, an increase in the amount of TV watched shows an increase in score." The description is classified as nominal because the relation between the vocabulary scores and age (namely, that vocabulary scores increase with age) is not explicitly described. Obviously, the nominal classification does not preclude the possibility that the participant mentally represented the order information.

2. Ordinal descriptions included the explicit mention of the relation between the third variable and the dependent variable. In the foregoing example, the order information might have been described as "And, the vocabulary scores for younger people are less than for older people."

3. Metric descriptions included a more detailed quantitative explanation of the relations between the third variable and the dependent variable. A description was classified as metric if it included description of the interval or ratio relation between the third variable and the dependent variable. That is, a metric description included information about changing the z - y relations as a function of the z -variable. For example, a metric description of the z -variable for the intercurve spacing graph in Figure 2b would have included a statement about the increasing influence of sleep on energy level as sleep level increases.

The visual inspection and description processes are temporally integrated, making it impossible (and undesirable) to separate out the time needed to verbally describe the graph from the time to inspect the graph. In any case, the total time spent on a graph appears not to be primarily a reflection of the length of the verbal description. To estimate the time needed to articulate a description, we recorded how long it took a separate group of 3 participants to read a typical nominal, ordinal, and metric description for the line and point perspectives, for both the sparse and dense conditions. We then weighted each description time by its frequency to derive a total speaking time for the four conditions. The descriptions took an average of 11.3 s to read, only one third of the time of an average trial, and there was almost no variance as a function of condition. Thus, the time to output the verbal description accounts for little of the difference among the perspective and density conditions. Keeping in mind that these descriptions were intimately intertwined and coordinated with the inspection,

rather than simply output at the end of the inspection period, their impact on the total time is likely to be even less than this analysis might suggest.

Results and Discussion

Time and gaze durations. The distribution, number, and duration of inspection gazes were highly consistent with the integrative model, which proposes that graph comprehension involves pattern encoding, retrieving or inferring quantitative relations, and interpreting the referents by the variable names, scales, and values. A typical inspection involved numerous gazes to different regions, such as from the pattern to a label or axis, back to the pattern, and back to the axis. Overall, there was an average of 33 gazes per trial, averaging 1.1 s each in duration, meaning that each of the five sectors had multiple gazes. The average number of gazes varied from 29 (totaling 33 s) for a relatively simple graph, the intercurve spacing graph, to 40.5 gazes (totaling 40.6 s) for the non-cross-over dense graph.

The cyclic nature of the integrative processes is clearly evident in a prototypical scan pattern, as depicted in Figure 5 for a dense version of a cross-over graph. This figure shows the graph and, superimposed on top, individual lines with arrowheads. Each line represents one transition between successive gazes on different regions. Most of the transitions are between the graph pattern and the x -, y -, and z -labels and the title; this results in the large number of horizontal and vertical lines leading into or out of the graph sector. In addition, there are a small number of transitions from one label sector to another. After reading the title, the participant spent a short time inspecting the graph, and then began slowly to describe the variables and their interrelations. The participant provided the following description, which was fairly typical of this type of graph:

This is vocabulary score vs. age by TV watched in hours . . . and it shows that vocabulary scores increase with age very dramatically for someone who watches a lot of TV and not so dramatically for someone who watches a little of TV.

The focus on the quantitative nature of the x - y

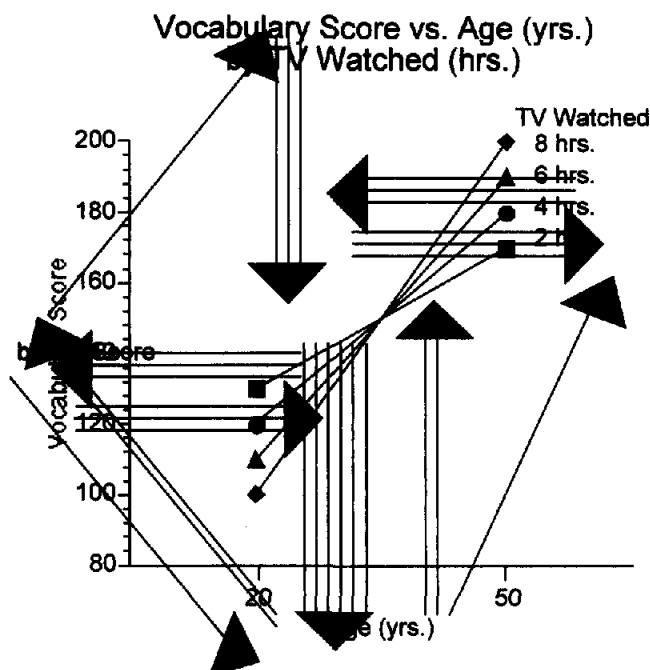


Figure 5. A diagram of the number of transitions between regions from a prototypical trial. Each transition is indicated by a line beginning from the center of the regions that were fixated. Note the large number of transitions between the graphic pattern and the x-axis, y-axis, and z-labels regions. This emergent pattern of gazes is consistent with the proposal that the information about the pattern itself is integrated with the interpretation of labels, scales, and units. The viewer, like most, focused on the main x-y functions, saying, "This is vocabulary score versus age by TV watched in hours. And it shows that vocabulary scores increase with age very dramatically for someone who watches a lot of TV and not so dramatically for someone who watches a little of TV."

relation, as in this description, is the typical interpretation of these graphs. The description appears to emerge from an inspection of the distinct functions that are then related to their referents in terms of the labels and values on the axes. Thus, the majority of gazes occur between the pattern region and the other sectors that contain the referent names, units, and numerical information. This inspection pattern, which is typical, suggests the need to refresh and integrate the information across the different regions of the graph.

Figure 6 shows the distribution of gaze times across the five sectors of the graph. As predicted by the integrative model, people spent considerable time on the sectors that provided labels and values of the variables, in addition to time on the pattern sector. Part of the time in the title sector (1-2 s) may be accounted for by the fact that

participants read the title at the beginning of each trial. Nevertheless, the distribution of gaze time across the other sectors clearly supports the integrative model and fails to support the hypothesis that people focus primarily on the graph pattern.

The overall time that people took to inspect a graph and generate a description typically showed effects of a graph's density and perspective, as indicated in Table 1. Consistent with the model, there was an increase in processing time if the density manipulation increased the number of distinct x-y functional relations (specifically, non-parallel lines), but not if the additional lines were parallel. In addition, there were marked differences as a function of the perspective of a graph, suggesting that participants derived different mental representations of the same data set, depending on the perspective. A prediction of the model

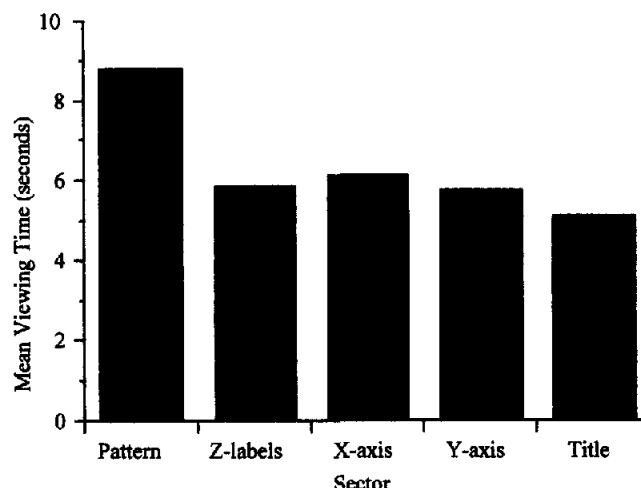


Figure 6. The total time spent on each of the five sectors of a graph, averaged across the participants and graphs in Experiment 1. Although the time on the pattern is almost 9 s, the remaining sectors each receive close to 6 s of gazes, and combined, they receive 24 s, which exceeds the time on the pattern region by a factor of more than 2.

was that the interpretation of dense versions would take more time, particularly if they depicted additional, unique functions. This hypothesis was supported, as shown in Figure 7. For the cross-over interaction data set, the 3.4 s interaction of density and perspective, $F(1, 32) = 5.82$, $p < .05$, was consistent with the expectation that processing time increased with the number of distinct x - y relations. Specifically, the addition of nonparallel lines for the non-cross-over graph led to a larger increase in the total time than the addition of nearly parallel lines for the cross-over graph. Similarly, for the common-value interac-

tion data set, the interpolation of additional lines for the line perspective was expected to have more effect than for the point perspective, and there was an insignificant interaction of density and perspective for the time measure $F(1, 32) = 2.58$, ns , but the analysis was significant ($p = .051$) for the number-of-gazes measure. For the acceleration data set, the additional interpolated lines were parallel to the existing lines in each case. Thus, as expected, there was no interaction between perspective and density, $F(1, 32) = 0.64$. Overall, the pattern of results for the total times supports the conclusion that the number of

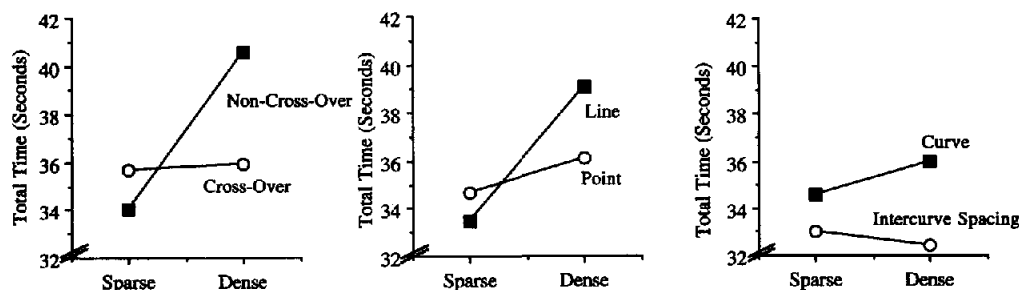


Figure 7. The total time to interpret the sparse and dense versions of the cross-over and non-cross-over perspectives (left), the line and point perspectives (middle), and the curve and intercurve spacing perspectives (right). Note that the density manipulation has its largest effect in the non-cross-over perspective case, when the additional lines have different slopes. It also has some effect for the line perspective, where the additional lines have intermediate slopes.

distinct functional relations influences the duration of the interpretation process.

Consistent with the integrative model of graph comprehension, the density and perspective frequently had significant effects on the time the person spent on the pattern and on the labels and axes. For example, the interaction of perspective and density for the cross-over interaction data set was general across most of the sectors, and significant in three of the five sectors. The interaction of density and perspective for the common-value interaction data set was confined to the pattern and *z*-label sectors, perhaps because these sectors are involved in identifying particular lines with values and extracting the additional ordinal and metric information. Finally, for the acceleration data set, in which the curve perspective took 2.6 s longer, $F(1, 32) = 4.42, p < .05$, in the total time measure, more time was spent for this perspective on the pattern and the *x*-axis sector. By contrast, for the *z*-labels, the intercurve spacing perspective took significantly longer.

The overall distribution of gaze time across the different sectors, the pattern of sequential gazes, and the total time and number of gazes are consistent with the hypothesis that graph comprehension involves cycles of pattern recognition and interpretation and integration. Limitations in the ability to interpret and integrate multiple functions may give rise to the decomposition into chunks and also contribute to the systematic lack of understanding the metric properties of the *z*-*y* relations, as described below.

Descriptions and comparison task performance. The relative ease of representing the *x*-*y* relations and the relative difficulty of inferring the metric properties of the *z*-*y* relations were apparent in the types of graph descriptions generated by the participants. Most often, the descriptions included metric information about the major *x*-*y* relations identified by the individual *z*-values, and typically they only included either nominal or ordinal information about the *z*-variable. The average proportions of nominal, ordinal, and metric descriptions of the *z*-*y* relations are given in the histograms in Figure 8. Clearly, the frequency of nonnominal descriptions varied for the different perspectives within a data set. As predicted, ordinal or metric descriptions of the *z*-*y* variable were most frequent for the curve and intercurve spacing perspectives,

which had parallel *x*-*y* lines. Ordinal and metric descriptions were the least frequent for the non-cross-over graph, which had *x*-*y* lines with the most dissimilar slopes. Also as predicted, the density of the data set influenced the descriptions, $\chi^2(1, 408) = 22.9, p < .01$. It was expected that if the dense version made the *z*-dimension more salient, it should increase the likelihood of viewers providing a quantitative interpretation of it. Consistent with this expectation, there were more ordinal and metric descriptions for the dense than for the sparse data sets (a total of 50.0% vs. 27.0%) and correspondingly fewer nominal descriptions. The only case in which the proportion of metric descriptions decreased with increasing density was for the intercurve spacing graph, where the interpolation manipulation apparently obscured the unequal spacing of the parallel lines.

The perspective-comparison task was used to assess whether the participants had an accurate representation of the *z*-*y* relations from the initial graph, and hence, whether they could accurately judge whether the alternative perspective was the same or different. Overall, participants were not very accurate in this task (52.2% correct), indicating that in general they did not have a good representation of the *z*-*y* relations in the initial graph. They were not simply guessing, however, because their accuracy was higher if the description had included metric information about the *z*-*y* relations. Specifically, accuracy was higher on trials for which participants had given metric descriptions (74.2%) than nominal and ordinal descriptions of the *z*-*y* relations (50.8% and 50.6%). Overall, the accuracy was slightly higher for the dense than for the sparse conditions, 55.9% and 48.5%, respectively, although this difference was not significant, $F(1, 32) = 1.58, ns$.

Summary. The time and number of gazes on the different graph types support an integrative model of graph comprehension. The processes of recognition, translation (which may involve considerable inferential processing), and identification are accompanied by gazes that reflect the processes by which the person relates the graphic function to the information specified in the labels, title, and axes. The interpretation of the graph emerges from incremental recognition and integrative processes, processes that appear to be severely limited. In fact, the limitations in integra-

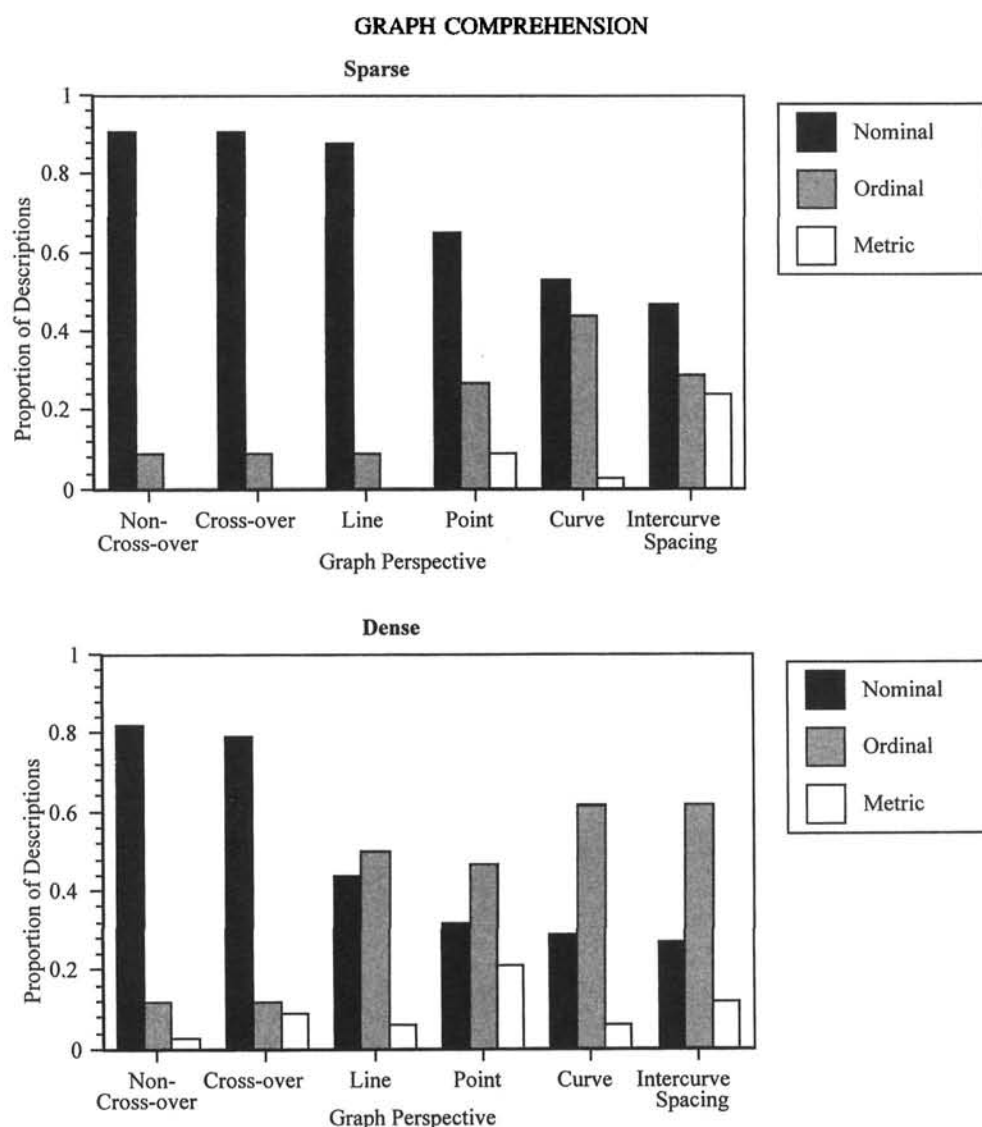


Figure 8. A histogram of the proportion of nominal, ordinal, and metric descriptions for each of the graph perspectives for the sparse and dense versions in Experiment 1. Most of the descriptions are nominal for the sparse versions. The proportion of ordinal descriptions increases with the density manipulation, although it has less effect for the graphs depicting the cross-over data set.

tion may lead to the emphasis on the x - y relations over the z - y relations and hence the systematic difficulty that people have in recognizing the equivalence of two alternative perspectives of the same data.

Experiment 2

The goal of this study is to provide a more precise characterization of the interpretation pro-

cess by fitting the eye-fixation patterns to an algebraic model of the processing account. Specifically, the model predicts that the number of gazes in each sector should be monotonically related to the number of distinct x - y functions; for simplicity, we assume that the relation is linear. The model predicts that for each additional x - y function, the person encodes the function, interprets it, and relates it to the various referents that are associated with the labels and values.

To examine more systematically the relationship between the pattern of eye fixations and the verbal description, we constrained the task by asking the person to answer a question about the effect of one variable on the dependent variable, either the x - y relations or the z - y relations. For example, the two types of questions for Figure 1a involving vocabulary scores would be, What happens to vocabulary scores as age increases? and What happens to vocabulary scores as the amount of TV watched increases? Although question answering may at first glance appear to be simple fact retrieval, it is an interpretive task. Thus, at some level, a person's comprehension processes are expected to be similar to those in Experiment 1. On the other hand, the task is more focused, and this constraint should lessen individual variation and thus allow us to test a simple quantitative version of the integrative model.

To examine the hypothesis about the influence of the number of functions on the interpretation process, we developed a coding scheme that algebraically expresses the number of distinct functional relations in the line graphs. This coding scheme is based on the hypothesis that a single line will evoke a single functional relation, such as "y is increasing . . ." or "y is decreasing . . ." (Details of the coding are given in the *Method* section.) Additional parallel lines do not add any additional information to this relation and therefore do not increase the count of the number of distinct functions. Lines that have the same sign but not the same magnitude are rated as having an additional one half of a distinct relation, reflecting the apparent variability in whether or not people encode these lines as distinct functions. Finally, an additional line with a slope of the opposite sign is coded as a distinct function. The functional relations are merely the first part of the coding to be generated. The person must also determine what "y" stands for; what "x" represents; the scale and magnitude of the relation; and by gradually building up these descriptions, arrive at a more integrated understanding of the graphic information.

This experiment also allows us to compare the pattern of gazes associated with answering different questions, either about the x - y relations or the z - y relations. In general, viewers emphasize the x - y relations over the z - y relations, and it appears to be more difficult to retrieve the z - y relations

(Shah & Carpenter, 1995). The x - y relations may be more salient for a variety of processing reasons. The x - y lines are more likely to automatically evoke retrieval of the ordinal and metric interpretations of these functions. By contrast, inferring z - y relations may require ordering and internally labeling the individual lines by viewing the z -labels and identifying the relative position of those lines on the y -axis. The necessity of more complex inferential processes will be reflected in additional time and numbers of gazes on the referential regions of the graph, particularly the z -labels and y -axis sectors.

In sum, we will focus on two types of analyses of the eye fixations in the question-answering task in Experiment 2. In one analysis, we will propose an algebraic model that relates the number of gazes in the different regions of the graph to the properties of the pattern. Second, we will compare the processing evoked by questions that address either the x - y or the z - y relations.

Method

Participants. Twenty-nine Carnegie Mellon University students participated, all of whom were native English speakers and had normal, uncorrected vision. The data for an additional 4 participants could not be analyzed because of an inability to record accurately their eye fixations.

Design and stimuli. The design and stimuli of this experiment were identical to those in Experiment 1, with two exceptions. First, there was no comparison task, and hence no comparison graphs were used. In addition, whether the question addressed the x - y relations or the z - y relations was an additional factor. The assignment of questions to graphs was designed so that each participant answered one x - y question and one z - y question for each of the six graph types, for a total of six x - y questions and six z - y questions. Specifically, each graph was randomly assigned to an x - y or a z - y question, with the constraint that one x - y and one z - y question occurred once for each graph type. The x - y and z - y questions were balanced by forming a second condition with the opposite assignment of questions to graph. Half of the participants were randomly assigned to be in one question condition and half to the other.

Procedure. The procedure was similar to that in Experiment 1, with a few changes to accommodate the differences in tasks. First, in the present experiment, the title of the graph was read by the experimenter before the trial began, and then the participant was given a specific question to answer about the graph. For the graph in Figure 1a, for example, the experimenter would ask one of two possible questions: (a) What happens to vocabulary score as age increases? or (b) What happens to vocabulary score as the amount of TV watched increases? After hearing the question, the person fixated a calibration point in the top-left corner of the screen and pushed a button to view the graph and answer the question. Thus, the gaze pattern and durations reflect the processes by which the participant formulated and gave the response.

As in Experiment 1, the session began with instructions that introduced three-variable graphs and the task of answering questions about the relationship between variables. After these instructions, the eye-tracking equipment was calibrated. The experimental session included 1 practice graph, 12 experimental graphs, and 5 interspersed filler graphs. The graphs were presented in one of two randomized orders. The first order was randomly determined, and the second order was its reverse. Participants were randomly assigned to one of two orders.

If in a trial a total of more than 15% of the observations were unclassifiable (due to noise or lost observations), the trial was not analyzed. Using this criterion, a total of 16 trials (from 12 different participants) were omitted, and the missing data for these conditions were estimated using the procedure for estimating missing data in Winer (1962).

The proportion of the total gaze duration that was not assigned to a region averaged 12% overall, which includes blinks, lost observations, saccades, and gazes that were less than the minimum duration of 250 ms. This proportion did not significantly vary between the two perspectives for each of the acceleration or common-value interaction data sets, nor as a function of question or complexity. For the cross-over interaction data set, the proportions varied only by a small amount, such that 11% of the total duration in the cross-over condition and 13% in the

non-cross-over condition were not apportioned to any regions.

The coding scheme. Each graph was assigned a value representing the number of unique x - y functions. A single line was counted as one function, and any additional parallel lines did not contribute to the count. Thus, for example, curve and intercurve spacing graphs, regardless of their density, were rated as having one functional relation. An additional line that had a slope with the same sign but different value was assigned half a point. The interpolation of additional lines in the same direction as the two extremes did not add to the count because the two extreme lines fully describe the data. Thus, both the sparse and dense version of point and cross-over graphs were rated as having 1.5 functional relations. A horizontal line, which falls between a "similar slope" and a unique functional relation was counted as .75 additional functions. For example, the sparse line graph was coded as 1.75 functions. Finally, a line with a slope that had an opposite value from the first line was counted as 1 additional function. Thus, the sparse non-cross-over graph was rated as having 2 functional relations. The dense version of the non-cross-over graph has 3 distinct functions (the original 2, plus $2 \times .5$ for similar lines in opposite directions). The number of distinct functional relations of the graphs ranged from 1 to 3.

Results and Discussion

The total time as well as the number and distribution of the inspection gazes in the question-answering task are consistent with an integrative model of graph comprehension. Overall, the total time was relatively long (averaging 23 s), indicating that the question-answering processes are fairly complex. Across the six graph types, the shortest average time was 17.8 s for the intercurve spacing graph, and the longest was 31 s for the non-cross-over graph.

Although the total time indicates the complexity of the processes, its integrative nature is more apparent in the distribution of gazes across the sectors, as shown in Figure 9. As in Experiment 1, the time on all of the labels and axes (the x -axis, z -labels, y -axis, and title) exceeds the time spent on the pattern, with the former four sectors receiving an average of 14.1 s inspection com-

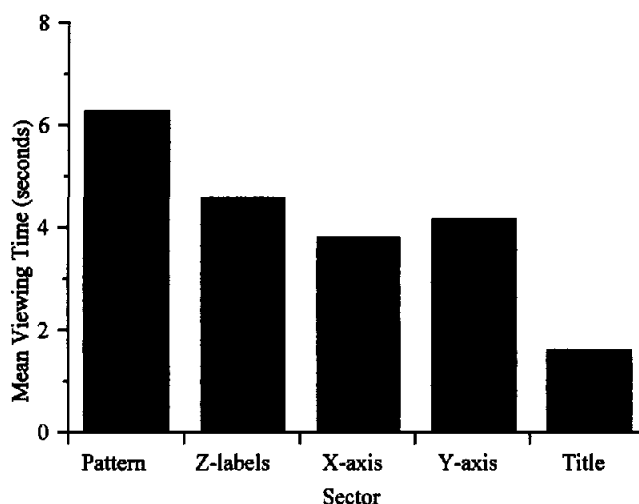


Figure 9. The total time spent on each of the five sectors of the graph, averaged across the participants and graphs in Experiment 2. As in Experiment 1, the combined time on z-labels, x-axis, y-axis, and title exceeds the time on the pattern. Note that the time on the title was less here than in Experiment 1, in part because the title was read by the experimenter before the start of a trial.

pared with 6.3 s on the pattern. The gaze duration on a sector was about 1 s, which is similar to what was found in Experiment 1. The number of gazes to different sectors averaged from 18.4 for the intercurve spacing perspective to 32.6 for the non-cross-over perspective, with an overall average of 23.5 gazes. Thus, the number of gazes also reflected the complexity of interpretation processes. In sum, the inspection duration and fixation patterns suggest that a large proportion of the time is consumed by linking interpretations derived from all of the various parts of the graph, even in the restricted task of question answering.

Comprehension of x - y relations. To more systematically analyze the interpretation of the x - y relations, we applied the simple algebraic model of the number of distinctive functions in the pattern in order to predict the distribution of gazes in the different regions of the graph, as well as the differences among the six graph types.

The distribution of the fixations across the different types of graphs is determined by two main processes in the comprehension of the x - y relations. In the typical inspection pattern, the person first scans the graph in order to find the location of the referents to the question. Second, the person identifies and comprehends a distinctive functional relation, which not only involves fixating the display, but also the axes and z-label.

The model can be described by the following steps: (a) Scan the graph to obtain an overview and locate the referents of the question (view each of the five regions once); (b) identify a unique function (view the pattern); (c) describe the nature of the x - y relation for that function: (1) Identify change (increase or decrease) in the x -axis for that function, including its direction, scale, units, and/or referent (view the x -axis, the pattern, and the x -axis), (2) identify change (increase or decrease) in the y -variable for that function, including its direction, scale, units, and/or referent (view the y -axis, the pattern, then the y -axis), (3) identify z-label in order to identify the direction, scale, unit and/or referent (view the z-labels, the pattern, then the z-labels again).

Steps (b) and (c) are applied repeatedly to different functions. The three steps listed under (c) can take place in any order, corresponding to the strategy that the person uses to comprehend the graph. For example, a person might interpret the graph by identifying a particular line by viewing the z-labels, then identifying the x - y relationship, resulting in a description like "For z1, as x increases y increases. . . ." Another person might describe an x - y relationship, then identify the line by its z value; in this case, the description may be, "As x increases, y increases for z1. . . ." In both cases, the participants identi-

Table 2
The Predicted and Average Number of Observed Gazes on the Five Sectors (Experiment 2)

Sector	Observed no.	Predicted no.	Correlation across the 12 graph types
Total	22.2	19.3	.96
Pattern	7.2	5.8	.87
X-axis	4.5	4.2	.77
Y-axis	4.2	4.2	.90
Z-labels	4.5	4.2	.88
Title	1.7	1.00	

fied the individual functions and their referents. Thus, the model predicts number of gazes in each region rather than the sequence of the gazes for each graph type.

A second assumption of the model is that in addition to scanning the graph, the person fixates each label twice, with the pattern inspection in between. The value of approximately two gazes per referential region or function was chosen in order to fit the data for graphs that were rated as having only one function, and it was found to predict the data for more complex graphs as well, although it tends to underestimate the number of gazes. The multiple inspections might reflect several subprocesses, such as confirming information, checking for new information, or refreshing their memory about the referential information.

These reasons are consistent with the assumption that there are severe limits to the amount of information that is encoded or retained in each gaze. However, at this time, this aspect of the model is primarily descriptive rather than a specification of the particular mechanisms. Overall, this model predicts the aggregate number of gazes in the different regions for specific graph types (as shown in Table 2). It also predicts the average total number of gazes to answer x-y questions, which is shown in Figure 10 along with the observed total number of gazes for the various perspectives.

As shown in Table 1, some of the underestimation between the predicted and observed number of gazes occurs for the pattern and title regions. The number of gazes on the pattern sector may be underestimated because the process of identifying a referent may occasionally require an additional confirmation step. The number of gazes on the title may be underpredicted because the identification of some variables can require viewing the title. The influence of the number of distinct functions on the interpretation process is supported by a high correlation between the predicted number of gazes (based on the formula) and the observed number of gazes across the 12 graph types, $r(10) = .96$, $p < .01$, as shown in Table 1. Furthermore, the correlation between the predicted and actual number of gazes in the

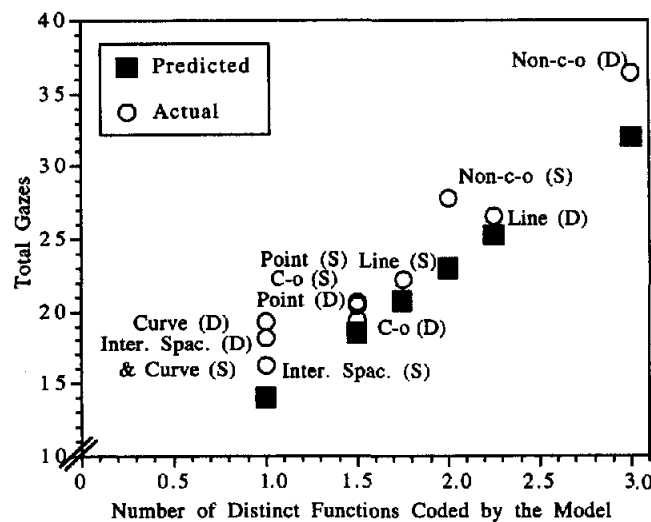


Figure 10. A graph showing the relation between the number of predicted total gazes using the algebraic model and the observed total gazes for the various graph perspectives.

different regions is also high, ranging from .77 to .90, although the model underestimates the number of gazes and oversimplifies the mapping between the processes and the gazes.

The number of distinct x - y functions is a better predictor of the number of gazes than the visual complexity of the graphs (as rated by an independent group of 10 students). A partial correlation between a prediction based on the number of distinct x - y functions and actual number of gazes is significant, even when the effect of rated visual complexity is partialled out, $r(10) = .97, p < .01$. By contrast, the correlation between the visual complexity ratings and the total number of gazes is not significant, $r(10) = .43, ns$, and the correlation is negative when the number of functional relations is partialled out, $r(10) = -.67$. Of course, visual complexity is in part a function of the number of distinct x - y functions, but to the extent that the two are conceptually separable, it would appear that the more predictive metric is the number of visual chunks to be interpreted.

Graph-type analysis. The demonstrated influence of the number of distinct functions on the interpretation of graphs implies that two graphs that depict the same data may evoke different internal representations through a different interpretation process. Thus, just as in the previous experiment, the overall time that people took to inspect a graph and generate a description typically showed effects of a graph's density and perspective. Table 3 provides the mean gaze durations for each sector for the x - y questions (omitting the comparable times for z -questions) and any F values greater than 2.5. (If an interaction term is not listed, no F value exceeded the cutoff.) The main pattern of results is similar to that for Experiment 1, as shown by the data in Figure 11. These graphs show the time participants took to answer x - y questions, which was the task closest to the general description in Experiment 1.

As expected, the addition of lines that increased the number of distinct x - y functional relations (specifically, nonparallel lines) was associated with increased time compared with the addition of lines that did not increase the number of distinct x - y functional relations. The addition of nonparallel lines to the non-cross-over perspective had a bigger effect than did the additional,

almost parallel lines for the cross-over perspective, $F(1,27) = 8.74, p < .05$, for both x - y and z - y questions. Similarly, the addition of lines to the line perspective graph had a greater effect than the addition of lines to the identical data in the point perspective for x - y questions. The overall effect of density by perspective for this data space, shown in Table 3, is not significant because the effect of density and graphic perspective was different for the two types of questions, as indicated by the three-way interaction of density, graph type, and question type, $F(1, 27) = 3.74, p < .05$. Finally, as in Experiment 1, the addition of lines to both the curve and intercurve spacing graphs had similar, negligible effects for both x - y and z - y questions. These results further support the model of graph interpretation described earlier, in which the number of unique x - y functions, not the graphic perspective or density of the data set per se, has the greatest influence on the interpretation processes.

Density manipulation. Supporting the hypothesis that the density of a graph will have an impact on the inspection process primarily when additional lines depict unique functional relations, the density manipulation did not have an overall influence on the time on the graph. The density manipulation did influence the time in the pattern and z -label sectors of the graph, perhaps because these sectors reflected the time needed to associate lines to particular quantitative values and to infer general trends. This effect was greatest for those graph types for which additional lines increased the number of functional relations. For the more complex conditions, the pattern sector had inspection times that were almost 2 s longer (7.4 vs. 5.1 s), $F(1, 27) = 4.66, p < .05$, and more gazes (8.1 vs. 6.3 gazes), and the z -labels received about 1.6 s more gazes (5.3 s vs. 3.7 s), $F(1, 27) = 5.00, p < .05$. The importance of the functional complexity of a graph is made more apparent in the description of the algebraic model of the comprehension of x - y relations described earlier.

The current experiment has three main conclusions. First, even in the restricted task of question-answering, the comprehension of a graph arises from reiterative applications of pattern recognition and interpretation processes. Second, the interpretation of x - y relations is slightly easier than the interpretation of z - y information, presum-

Table 3
*Mean Viewing Time (in Seconds) to Answer Questions About X
 and Large F Values (Experiment 2)*

Graph type	Total	Pattern	X axis	Z label	Y axis	Title
The cross-over interaction data set						
Non-cross, sparse	24.87	5.59	4.34	4.00	4.84	2.95
Non-cross, dense	35.20	11.92	5.53	6.18	5.97	2.04
Cross-over, sparse	19.59	5.85	3.82	3.73	2.67	1.49
Cross-over, dense	22.15	9.48	3.54	2.96	3.57	0.71
Perspective	16.74**§	—	5.09*§	11.56**§	16.09**§	8.30**§
Question	—	—	8.32**	10.23**§	—	—
Density	—	6.73§	—	—	—	—
Perspective × Question	—	—	—	—	—	5.33*
Perspective × Density	8.74**§	5.37*§§	13.23**§	—	—	—
Question × Density	—	—	4.41*	—	—	—
The common-value interaction data set						
Line, sparse	22.14	4.85	5.36	3.34	3.61	2.39
Line, dense	26.87	7.42	4.09	5.92	4.60	2.07
Point, sparse	19.64	4.21	3.63	3.01	4.28	2.10
Point, dense	21.36	4.87	4.47	4.63	3.34	1.30
Perspective	4.34*§	9.37**§	—	—	—	4.59*
Question	—	—	—	—	—	4.12
Density	—	—	—	4.92*	—	—
Perspective × Density	—	—	7.94**§	—	—	—
Perspective × Density × Question	3.35§	—	—	—	4.17§	—
The acceleration data set						
Curve, sparse	18.78	5.88	4.19	1.68	3.75	1.05
Curve, dense	20.49	7.18	4.31	2.85	2.80	1.13
Intercurve, sparse	15.82	3.69	3.79	1.78	2.87	1.87
Intercurve, dense	16.66	4.84	3.22	2.74	2.72	1.17
Perspective	9.26**	11.39**	—	—	13.56**	—
Question	5.98*§	—	—	29.58**§	11.11**§	—
Density	—	—	—	8.28§	—	—
Question × Density	—	—	—	—	4.34*	—
Perspective × Density × Question	—	—	—	—	—	3.81

Note. Dashes indicate that the associated *F* statistic was small.

p* < .05, gaze-duration effect significant. *p* < .01, gaze-duration effect significant.

§*p* < .05, number-of-gazes effect significant. §§*p* < .01, number-of-gazes effect significant.

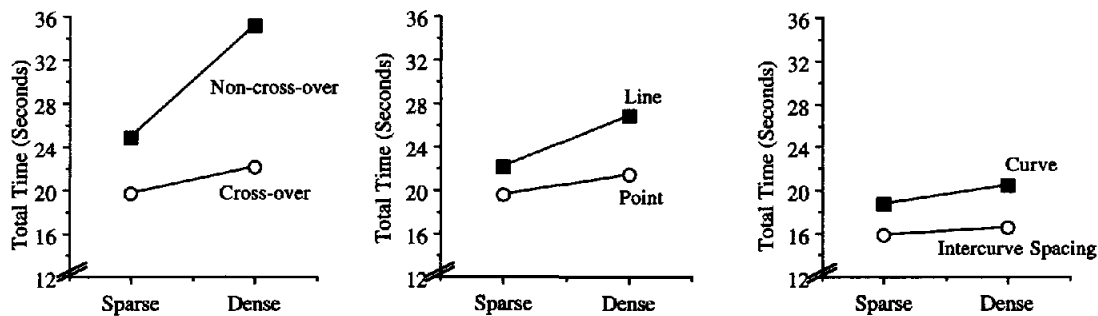


Figure 11. The total time to answer x - y questions about the sparse and dense versions of the cross-over and non-cross-over perspectives (left), the line and point perspectives (middle), and the curve and intercurve spacing perspectives (right). As in Experiment 1, the dense version takes more time in the conditions in which the additional lines have different slopes, namely the non-cross-over and line perspectives.

ably because of the ease of retrieval of the more familiar interpretations of x - y functions compared with the inferences required to interpret the z - y relations. Third, graph interpretation is influenced by the number of distinct functions in the graph. Two graphs that depict the same data but differ in the number of distinct functions that are depicted may lead to different internal representations.

General Discussion

The results from these studies support an integrative model of graph comprehension in which people incrementally encode and interpret the main functional relations depicted in a graph. According to this model, graph comprehension involves identifying the main quantitative information in the graph, generally the x - y relations, and relating each individual function to its graphic referents, regardless of whether those referents have been previously identified. The time it takes to interpret a graph is highly related to the number of unique quantitative relations depicted in a graph, functions that must be individually interpreted and integrated into a coherent interpretation.

This model has three main features that are supported by the characteristics of the person's descriptions and eye-fixation patterns. First, the integrative model assumes that a major component of the interpretation process is relating graphic features to their referents and interpreting

the referents from different parts of the display. Indeed, the majority of time spent in graph comprehension involves reading and rereading information from the axes and label regions of the graph (x , y , and z), and less time is spent solely on the pattern of lines on the graph. The fact that people often reexamine different parts of the graph suggests that they are unable to keep track of the information available in the axes and labels.

Second, the comprehension process is incremental. People may at some point identify each individual quantitative function (which may include parallel lines) and relate each function to its graphic referents and to the evolving larger representation, resulting in numerous gazes on the axes and labels in addition to gazes on the graphic pattern. This conclusion is consistent with individual inspection patterns and is further supported by the fact that the number of unique x - y functions correlates with the number of gazes required to describe different graph types, as well as the distribution of gazes in the different regions of the graphs.

Finally, the model allows for a continuum of cognitive effort in order to comprehend the information in a line graph, depending on the complexity and number the graphic patterns and presumably also the familiarity of the referents and relations. In the current study, graphs that required a large number of inferences concerning the z - y relations took more time than those for which the z - y relations were more transparent. Of

course, the results of the comparison task in Experiment 1 indicated that people often had an incomplete representation of the z - y relations, which is presumably a reflection of the effort that would have been required to comprehend these relations more completely.

Levels of Integration

The current analysis uses a fairly macro level of analysis, emphasizing the integration of information across the major regions of the graph, such as the title and the graphic pattern. The grain size of the analysis is compatible with the assumption that these processes build on the more micro-level pattern recognition processes that have been described for simpler tasks and graphs. For example, Simkin and Hastie (1987) proposed several elementary pattern-encoding processes, such as anchoring, scanning, and superimposition of the visual features, in order to make perceptual judgments and comparisons. These processes are likely to occur in the current tasks, albeit at a finer level than our analysis emphasizes.

A second theoretical implication of the current model is that there should be an effect of the person's familiarity with the interpretation of a visual pattern; in other words, there should be an influence of individual differences in graphic knowledge. Consistent with this view, in other ongoing research, we have found that college students with little familiarity with graphs have enormous difficulty interpreting the types of graphs used here. Indeed, their difficulties parallel the problems of poor readers in the sense that they take a long time, rely on general knowledge, and make increasingly more mistakes as the complexity of the graph increases. Thus, although the model has focused on stimulus and task factors, it leaves conceptual room for the other half of the equation, which is the person's knowledge and skills. As a consequence, one way to simplify the graph interpretation process would be to train the intended audience so that they become familiar with multiline x - y patterns. The implication of this model is that individual differences in graphic knowledge should play as large a role in the comprehension process as does variation in the properties of the graph itself.

Practical Implications for Graphic Design

The major premises of the integrative model have implications for graph design. If identifying and tracking the referents is such a large component of the comprehension process, it should be useful to minimize its difficulty by the use of various graphic features. One such feature, often made in graphic design books, is to label lines directly and avoid the use of legends or keys (Kosslyn, 1994; Schmid, 1983; Shah & Carpenter, 1995). Identifying and tracking a referent may also be aided by the use of symbols or features that are already associated with a particular referent or value (Brockmann, 1991; Fisher, 1982; Mittal, Roth, Moore, Mattis, & Carenini, 1995; Roth & Mattis, 1990; Wickens, 1992). For example, blue and red colors could be used to represent cold and hot temperatures (Brockmann, 1991). Of course, this practice is limited and even could be problematic if the graph reader and graph designer have different prior associations of visual feature to referents. For example, the color green means "profitable" to financial managers but "infected" to health care workers (Brockmann, 1991). Overall, the suggestions often provided for minimizing the effort to identify graphic referents are consistent with a principle that Wickens proposed: the general proximity compatibility principle (Wickens, 1992). It states that the best graphical feature for a particular type of quantitative information is one that represents that information most explicitly. Although a useful rule of thumb, this principle provides no mechanism for resolving the conflicting factors that might influence a complex graph's interpretation.

Such conflicting factors arise in the display of higher dimensional data, such as three or more dimensions. An example of such a conflict was apparent in the line graphs used in the current study. The results showed that by their nature the x - y relations are more salient than relations involving the z -dimension, which require the reader to make inferences. Presumably, the same factors would operate if there were additional graphs depicting additional dimensions. Also, the current model argues against the practice of depicting higher dimensional data with the use of arbitrary dimensions in complex figures, such as the Chernoff faces (Chernoff, 1973); glyphs

(Anderson, 1960); or stars, ships, castles, and trees (Kleiner & Hartigan, 1981). Such mappings impose an even larger burden on identifying referents and values of the dimensions (Gower & Digby, 1981) and on abstracting quantitative trends across variables (Fienberg, 1979).

A decision about how to resolve the conflict among the salience of the depicted relations in a multivariable graph may depend on whether the goal of the graph is communication or data exploration. For example, it may be acceptable to depict only the minimal number of variables necessary to express a particular point in one graph, and to provide readers with captions or descriptions that help them identify the important quantitative features and their referents (Mittal et al., 1995). Such a solution is impractical when graphs are used to present "all of the data." For low-dimensional data sets, another improvement might be the use of animation, which at least provides continuity of the labels (Becker, Cleveland, & Wilks, 1988; Huber, 1987; Mackinlay, Robertson, & Card, 1991; Stuetzle, 1987). By contrast, as the current studies suggest, multiple perspectives of the same data presented statically require that the person reidentify the labels for each view. However, it should not be assumed that animation of a complex display will be interpretable automatically. Studies of how people comprehend animated displays of mechanical devices indicate that there are severe limitations on a novice's ability to track multiple parts of display simultaneously (Carpenter & Just, 1992). Hence, animation per se does not guarantee a circumvention of the person's processing difficulties.

The Semantic Content of the Graph

Although the current studies have emphasized the influence of a graph's visual characteristics on its interpretation, the model opens a way to understand the possible influence of the graph's semantics. Such influences have been documented by studies showing that people, particularly those who are relative novices at interpreting graphs, make more errors if the depicted relations are the reverse of a familiar relation (Shah, 1995; see also Gattis & Holyoak, 1996). Less skilled college students are more likely to describe a graph as conforming to their prior

knowledge than to following the syntax of the graph. For example, in interpreting a line graph with three variables (y as a function of x and z), a less skilled student is much more likely than a more skilled student to describe the x and z variables as causally related if the variables have a prior semantic relation (such as car accidents and alcohol consumption), where skill is operationalized as number and type of prior college math courses (Shah, 1995). Also, young children make more errors interpreting graphs that depict unfamiliar or complex variables than familiar and unidirectional variables such as time (Janvier, 1983). Finally, young children sometimes believe that graphs represent literal pictures of situations rather than abstract quantitative information (e.g., Janvier, 1981; Leinhardt, Zaslavsky, & Stein, 1990; McDermott, Rosenquist, & vanZee, 1987; Preece, 1990). That such errors are more evident with novices and children indicates that one outcome of learning to read graphs is that it enables the viewer to rely less on their prior knowledge of the typical relations among the variables and to get more information from the graph itself.

Summary

The results from these experiments suggest that the interpretation of a graph involves multiple, integrated cycles of identifying x - y functions and the variables associated with them. These integrative, reiterative processes form the heart of graph comprehension, even for relatively restricted tasks such as answering questions. Visual-recognition processes, such as encoding a line as having a positive or negative slope, are important component operations, but they are integrated with interpretive processes and, by themselves, do not consume the majority of time needed to understand these multiline graphs.

These experiments, which show that some graphs are easier and faster to interpret than other graphs, and some interpretation tasks are easier than other tasks, are consistent with the claim that the comprehension of visual displays tends to be easiest when simple pattern identification processes are substituted for complex cognitive processes (Bertin, 1983; Casner, 1990; Casner & Larkin, 1989; Wickens, 1992). These results further suggest, however, that even relatively

simple graphical displays require relatively complex cognitive processes.

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